

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Impact of Content Recommendation Systems on User Polarization

submitted in fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Impact of Content Recommendation Systems on User Polarization** ” in fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Shikha Tuteja. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

Social media platforms widely use content recommendation systems to personalise user feeds and improve user engagement. These recommendation systems analyse user interactions such as likes, comments, shares, and viewing behaviour to suggest relevant content. Although personalised recommendations increase user activity and retention, they can also influence user opinions and contribute to the formation of echo chambers and online polarization.

The objective of this project is to analyse the impact of different content recommendation strategies on user polarization using social media data collected from platforms such as Reddit and Twitter. The project involves data collection, preprocessing, sentiment analysis, and comparative evaluation of recommendation approaches. During preprocessing, unwanted information such as URLs, special characters, and stop words was removed using Natural Language Processing (NLP) techniques. Sentiment analysis was then performed to identify positive, negative, and neutral opinions expressed by users.

Two recommendation approaches were studied in this project: engagement-based recommendation systems and diversity-based recommendation systems. The engagement-based model recommends content based on user interaction metrics such as likes and shares, while the diversity-based model focuses on exposing users to a wider variety of viewpoints. Polarization metrics including sentiment variance and average sentiment score were used to compare the behaviour of both models.

The results of the analysis show that engagement-based recommendation systems tend to strengthen existing user beliefs and increase polarization, whereas diversity-based recommendation systems help reduce polarization by promoting exposure to diverse content. The project highlights the importance of designing balanced recommendation systems that can improve user experience while also reducing the risk of echo chambers and biased information exposure.

Keywords: Content Recommendation Systems, User Polarization, Echo Chambers, Sentiment Analysis, Social Media, Machine Learning, Natural Language Processing.

1. INTRODUCTION

1.1 Background

With the rapid growth of the internet and social media platforms, digital communication has become an important part of everyday life. Platforms such as Reddit, Twitter, Facebook, YouTube, and Instagram are widely used for sharing opinions, consuming information, and interacting with online communities. Due to the enormous amount of content generated every day, these platforms rely heavily on content recommendation systems to personalise user experience and improve user engagement.

Recommendation systems analyse user behaviour such as likes, comments, shares, watch history, and browsing patterns to recommend content that matches individual interests and preferences. Machine learning algorithms and data analysis techniques are commonly used to predict the type of content users are most likely to interact with. These personalised recommendations help platforms increase user retention and viewing time by continuously presenting relevant content to users.

Although recommendation systems improve user experience and platform engagement, they also introduce several social and ethical concerns. Most recommendation systems are designed to maximise engagement by promoting content that users are more likely to interact with. As a result, users are repeatedly exposed to similar viewpoints and opinions, which may gradually strengthen existing beliefs and reduce exposure to alternative perspectives. This phenomenon often leads to the formation of echo chambers, where users mainly consume information that supports their existing opinions.

Over time, echo chambers can contribute to user polarization by increasing differences in opinions among online communities. Polarization on social media platforms has become a major concern because it can influence public opinion, political discussions, and information consumption behaviour. To address this issue, some modern platforms have started exploring diversity-based recommendation approaches that expose users to a broader range of viewpoints instead of recommending only highly engaging content.

This project focuses on analysing the impact of different recommendation strategies on user polarization. By studying social media interaction data and applying sentiment analysis techniques, the project compares engagement-based and diversity-based recommendation systems to understand how they influence user opinions and online behaviour.

1.2 Problem Statement

Content recommendation systems used by social media platforms are primarily designed to maximise user engagement and retention. While these systems successfully improve user interaction, they can unintentionally encourage the formation of echo chambers by repeatedly recommending similar types of content. Continuous exposure to like-minded opinions may increase user polarization and reduce exposure to diverse perspectives.

There is a need to study how different recommendation approaches influence user behaviour and opinion formation. Understanding the relationship between recommendation systems and user polarization can help in designing balanced recommendation models that improve user engagement while also promoting informational diversity and reducing algorithmic bias.

1.3 Objectives

The main objectives of this project are:

- To analyse the impact of content recommendation systems on user polarization.
- To collect and preprocess social media data from platforms such as Reddit and Twitter.
- To perform sentiment analysis on user-generated content using Natural Language Processing (NLP) techniques.
- To compare engagement-based and diversity-based recommendation strategies.
- To evaluate polarization using metrics such as sentiment variance and average sentiment score.
- To study how recommendation systems contribute to the formation of echo chambers.

1.4 Scope and Limitations

This project focuses on analysing the effect of recommendation systems using publicly available social media datasets obtained from online sources such as Kaggle. The study mainly examines user opinions and behavioural patterns through sentiment analysis and recommendation simulations.

The project is limited to textual social media data collected from platforms like Reddit and Twitter. The recommendation models used in the study are simplified simulations and may not fully represent the complex algorithms used by real-world social media platforms. Additionally, the analysis is based on available datasets and does not include real-time user interactions or private platform data.

2. METHODOLOGY

The methodology of this project focuses on analysing the impact of different content recommendation systems on user polarization using social media data. The complete workflow includes data collection, data preprocessing, sentiment analysis, recommendation simulation, and evaluation of polarization metrics. The methodology was implemented using Python and various data analysis libraries.

2.1 Data Collection

The dataset used in this project was collected from publicly available online sources such as Kaggle. The dataset contains user-generated social media content from platforms including Reddit and Twitter. The collected data includes post text, user interactions, timestamps, likes, comments, and other engagement-related information.

The dataset contains approximately 50,000 posts and interactions from around 10,000 users. This large dataset helps in analysing user behaviour, opinion patterns, and interaction trends across different recommendation approaches.

2.2 Data Preprocessing

Before analysis, the collected dataset was cleaned and preprocessed to improve data quality and remove unnecessary information. Missing values and duplicate records were removed from the dataset to maintain consistency.

Several Natural Language Processing (NLP) preprocessing techniques were applied to the textual data, including:

- Conversion of text into lowercase
- Removal of URLs and special characters
- Removal of stop words
- Tokenisation of text
- Basic text normalization

These preprocessing steps help prepare the textual data for sentiment analysis and improve the accuracy of the results.

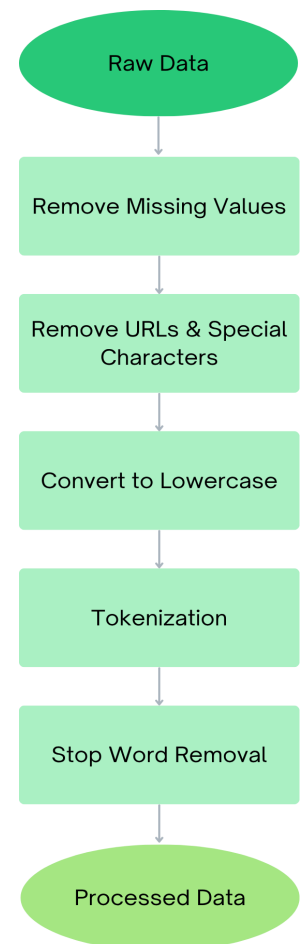


Figure 1: Data Preprocessing Workflow

2.3 Sentiment Analysis

Sentiment analysis was performed to determine users' opinions based on the content of their social media posts. Each post was classified into one of three sentiment categories:

- Positive
- Negative
- Neutral

The sentiment scores obtained from the analysis were used to understand how user opinions vary under different recommendation systems. Sentiment analysis also helped measure the level of polarization among users.

Python libraries and NLP techniques were used to process textual data and calculate sentiment values for each post.

2.4 Recommendation Simulation

To analyse the influence of recommendation systems on user behaviour, two different recommendation approaches were simulated in this project.

2.4.1 Engagement-Based Recommendation System

The engagement-based recommendation system prioritises content with higher user interaction such as likes, comments, shares, and viewing activity. This type of recommendation system is commonly used by modern social media platforms to maximise user engagement and retention.

However, repeatedly recommending similar content may reinforce existing beliefs and contribute to the formation of echo chambers.

2.4.2 Diversity-Based Recommendation System

The diversity-based recommendation system focuses on exposing users to a wider variety of content and viewpoints. Instead of recommending only highly engaging posts, this approach introduces diverse opinions and perspectives into users' feeds.

The purpose of this strategy is to reduce user polarization and encourage balanced information consumption.

Engagement-Based	Diversity-Based
Similar content	Diverse content
Higher engagement	Balanced exposure
Echo chambers	Reduced polarization

Table 1: Recommendation System Comparison Diagram

2.5 Polarization Metrics

To evaluate the impact of recommendation systems on user polarization, different statistical metrics were analysed. The primary metrics used in this project include:

- Sentiment Variance
- Average Sentiment Score

A higher sentiment variance indicates greater polarization among users, while lower variance represents more balanced opinions. The calculated metrics were compared for both engagement-based and diversity-based recommendation systems.

Python libraries such as Pandas and Matplotlib were used for data analysis and visualisation of results.

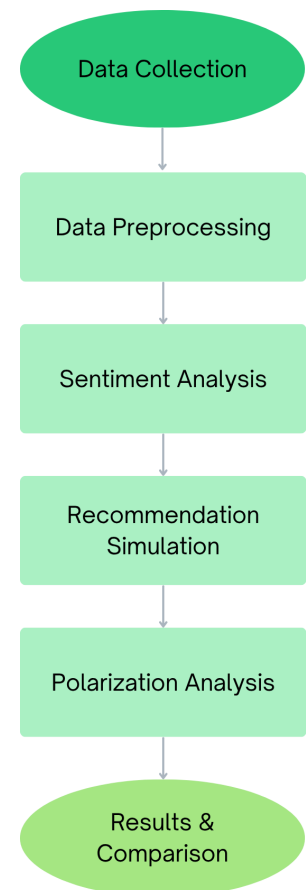


Figure 2: Methodology Flowchart

3. TOOLS AND TECHNOLOGIES

This project was implemented using various programming tools, libraries, and technologies for data processing, sentiment analysis, recommendation simulation, and visualization. The selected technologies helped in efficiently analysing social media data and evaluating the impact of recommendation systems on user polarization.

3.1 Programming Language

Python

Python was used as the primary programming language for implementing the complete project. It provides extensive support for data analysis, machine learning, Natural Language Processing (NLP), and visualization through various open-source libraries. Python's simplicity and powerful ecosystem make it highly suitable for research-based data analysis projects.

3.2 Libraries and Frameworks

3.2.1 Pandas

Pandas was used for data loading, cleaning, preprocessing, and manipulation of structured datasets. It helped in handling large amounts of social media data efficiently using DataFrames.

Purpose:

- Reading datasets
- Removing missing values
- Data preprocessing
- Data analysis

3.2.2 NumPy

NumPy was used for numerical computations and mathematical operations on arrays and datasets.

Purpose:

- Numerical calculations
- Statistical operations
- Array handling

3.2.3 NLTK (Natural Language Toolkit)

NLTK was used for Natural Language Processing tasks such as tokenization, stop-word removal, and text preprocessing.

Purpose:

- Tokenization
- Text cleaning
- Stop-word removal
- NLP preprocessing

3.2.4 TextBlob / VADER Sentiment Analyzer

Sentiment analysis libraries such as TextBlob or VADER were used to determine whether user posts expressed positive, negative, or neutral opinions.

Purpose:

- Sentiment classification
- Opinion analysis
- Sentiment score generation

3.2.5 Matplotlib

Matplotlib was used to create graphs, charts, and visual representations of the analysis results.

Purpose:

- Bar charts
- Pie charts
- Data visualization
- Comparative graphs

3.2.6 Seaborn

Seaborn was used to generate advanced statistical visualizations and improve the appearance of graphs.

Purpose:

- Statistical visualization
- Enhanced graph styling
- Distribution plots

3.3 Development Environment

Jupyter Notebook

Jupyter Notebook was used as the primary development environment for implementing the project. It allowed interactive code execution, visualization, and step-by-step analysis of data.

Advantages:

- Easy testing and debugging
- Interactive visualization
- Better documentation of code and results

3.4 Dataset Source

The dataset used in this project was obtained from publicly available online platforms such as Kaggle. The dataset contains social media interaction data from platforms like Reddit and Twitter.

4. IMPLEMENTATION

The implementation phase of the project focuses on analysing social media data to study the effect of recommendation systems on user polarization. The complete implementation was carried out using Python in a Jupyter Notebook environment. The workflow includes dataset loading, preprocessing, sentiment analysis, recommendation simulation, and visualization of results.

4.1 System Architecture

The system follows a sequential pipeline where raw social media data is collected and processed before performing sentiment analysis and recommendation-based evaluation. The implementation is entirely software-based and does not require real-time interaction with social media platforms.

The architecture begins with dataset collection and preprocessing, followed by sentiment analysis of user posts. After obtaining sentiment scores, two recommendation models—engagement-based and diversity-based—are simulated and compared using polarization metrics.

4.2 Dataset Loading and Preprocessing

The dataset was loaded into a Pandas DataFrame for analysis. After loading the dataset, several preprocessing operations were performed to improve data quality and prepare the textual data for sentiment analysis.

The preprocessing steps included:

- Removing missing values
- Removing duplicate records
- Converting text to lowercase
- Removing URLs and special characters
- Tokenization
- Stop-word removal

These operations helped in reducing noise from the dataset and improving the accuracy of sentiment classification.

4.3 Sentiment Analysis Implementation

Sentiment analysis was performed on user-generated posts to determine the emotional tone of the content. Each post was classified as positive, negative, or neutral using sentiment analysis libraries such as TextBlob or VADER Sentiment Analyzer.

The sentiment scores generated during this process were used to analyse opinion distribution among users. These sentiment values later helped in calculating polarization metrics for different recommendation approaches.

4.4 Recommendation System Simulation

Two recommendation strategies were simulated in the implementation phase to analyse their effect on user polarization.

4.4.1 Engagement-Based Recommendation

The engagement-based recommendation model prioritises posts with high user interaction such as likes, comments, and shares. Posts with greater engagement are more likely to appear repeatedly in users' feeds.

This simulation demonstrated how users are continuously exposed to similar opinions and content categories, which can strengthen existing beliefs and contribute to echo chamber formation.

4.4.2 Diversity-Based Recommendation

The diversity-based recommendation model selects content from different categories and viewpoints instead of relying only on engagement metrics. This approach increases exposure to diverse opinions and reduces content repetition.

The simulation helped compare how balanced recommendations influence sentiment distribution and reduce user polarization.

4.5 Polarization Analysis

After recommendation simulation, polarization metrics were calculated using sentiment scores. The main metrics used were:

- Sentiment Variance
- Average Sentiment Score

Higher sentiment variance indicated stronger polarization among users, while lower variance represented balanced opinions.

The results of both recommendation systems were compared to analyse their impact on user behaviour and opinion formation.

Model	Sentiment Variance	Average Sentiment
Engagement-Based	0.65	0.20
Diversity-Based	0.38	0.10

Figure 3: Polarization Metrics Comparison

4.6 Data Visualization

Various graphs and charts were generated using Matplotlib and Seaborn to better understand sentiment distribution and recommendation behaviour.

The visualizations included:

- Sentiment distribution charts
- Recommendation comparison graphs
- Polarization metric comparison
- User opinion analysis

These visualizations helped in interpreting the analysis results more effectively.

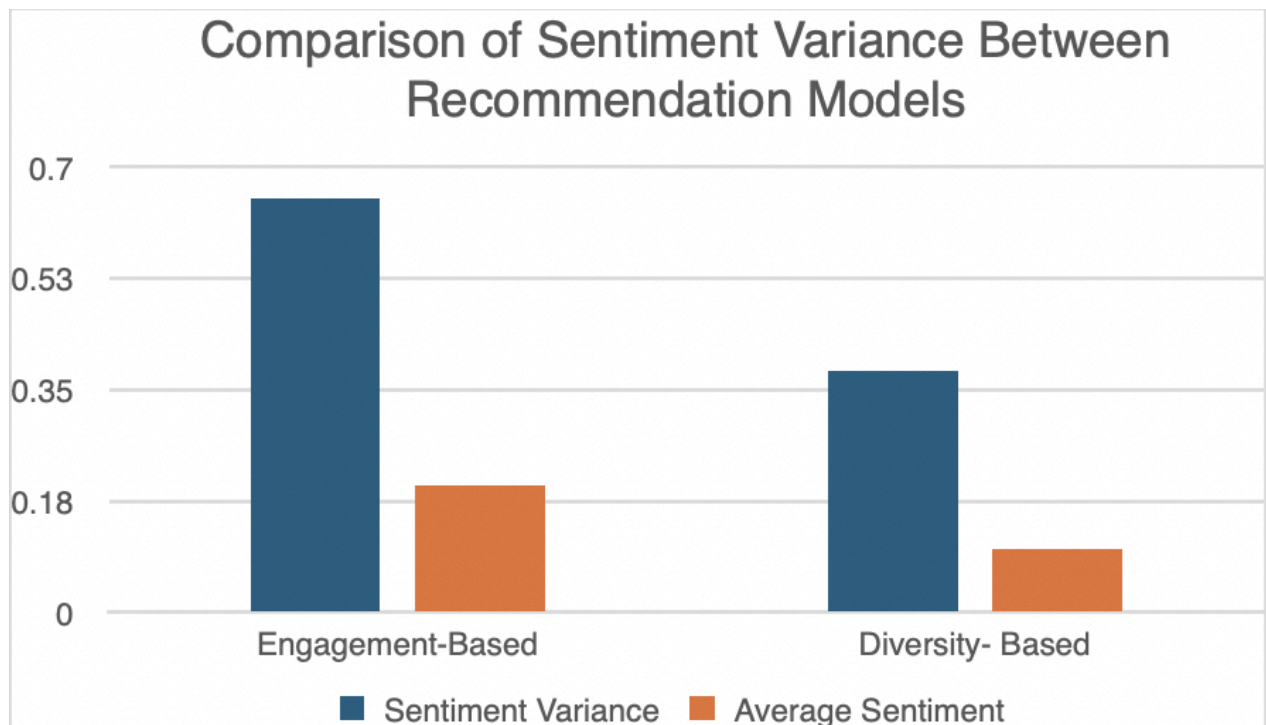


Figure 4: Final Visualization Outputs

5. MAJOR FINDINGS / OUTCOMES / RESULTS

This project analysed the impact of engagement-based and diversity-based recommendation systems on user polarization using social media data collected from platforms such as Reddit and Twitter. The analysis was performed using sentiment analysis techniques and recommendation simulations. The obtained results highlight how recommendation strategies influence user opinions, information exposure, and echo chamber formation.

5.1 Sentiment Analysis Results

Sentiment analysis was conducted on user-generated posts to classify opinions into positive, negative, and neutral categories. The analysis showed that user opinions varied significantly depending on the type of recommended content they were exposed to.

Posts recommended through engagement-based systems often contained emotionally aligned and opinion-reinforcing content, while diversity-based recommendations exposed users to a broader range of viewpoints and discussions.

Key Observations:

- Engagement-based recommendations increased repeated exposure to similar opinions.
- Diversity-based recommendations provided balanced exposure to different viewpoints.
- Sentiment distribution became more polarized in engagement-focused systems.

5.2 Comparison of Recommendation Models

The project compared two recommendation strategies:

- Engagement-Based Recommendation System
- Diversity-Based Recommendation System

The engagement-based recommendation system prioritised content with higher likes, comments, and interactions. This resulted in users repeatedly viewing similar content aligned with their interests and beliefs.

The diversity-based recommendation system introduced varied content into users' feeds, reducing repetitive exposure and promoting informational diversity.

The comparison showed that diversity-based recommendations helped reduce user polarization when compared to engagement-focused systems.

5.3 Polarization Metrics

To measure the level of user polarization, sentiment variance and average sentiment scores were calculated for both recommendation approaches.

The engagement-based recommendation system produced higher sentiment variance, indicating stronger polarization among users. In contrast, the diversity-based recommendation system showed lower variance, representing more balanced opinions and reduced echo chamber effects.

Model	Sentiment Variance	Average Sentiment
Engagement-Based	0.65	0.20
Diversity-Based	0.38	0.10

Figure 5: Polarisation metrics comparison

The results clearly demonstrate that engagement-based recommendation systems are more likely to reinforce existing user beliefs and increase polarization.

5.4 Echo Chamber Formation

The analysis indicates that users exposed primarily to engagement-based recommendations are more likely to interact with content that supports their existing opinions. Over time, this repeated exposure contributes to the formation of echo chambers where users receive limited exposure to opposing viewpoints.

On the other hand, diversity-based recommendation systems encourage users to explore different perspectives and reduce ideological isolation.

Key Findings:

- Engagement-based systems strengthen confirmation bias.
- Diversity-aware systems promote informational balance.
- Echo chambers are more prominent in engagement-focused models.

5.5 Overall Outcomes

The project successfully demonstrated how recommendation algorithms influence user behaviour and online opinion formation. The findings suggest that recommendation systems designed only for maximizing engagement may unintentionally contribute to polarization and biased information consumption.

The study highlights the importance of balancing engagement and diversity while designing recommendation systems for social media platforms.

Final Outcomes:

- Successfully analysed social media recommendation behaviour.
- Implemented sentiment analysis using NLP techniques.
- Compared engagement-based and diversity-based recommendation systems.
- Measured polarization using statistical metrics.
- Identified the role of recommendation systems in echo chamber formation.

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This project analysed the impact of content recommendation systems on user polarization using social media data collected from platforms such as Reddit and Twitter. The study focused on understanding how different recommendation strategies influence user opinions, engagement behaviour, and the formation of echo chambers.

The implementation involved data collection, preprocessing, sentiment analysis, recommendation simulation, and polarization evaluation using statistical metrics. Natural Language Processing (NLP) techniques were used to process user-generated textual content and determine sentiment patterns across different recommendation models.

The results showed that engagement-based recommendation systems tend to repeatedly expose users to similar viewpoints, which increases user polarization and strengthens echo chambers. In contrast, diversity-based recommendation systems encourage exposure to varied opinions and reduce the level of polarization among users.

The project demonstrates that recommendation algorithms significantly influence online information consumption and user behaviour. While engagement-focused systems improve user retention and interaction, excessive personalization can negatively impact informational diversity. Therefore, balancing user engagement with content diversity is important for designing responsible and unbiased recommendation systems.

Overall, the project successfully achieved its objectives by comparing recommendation strategies, analysing user sentiment, and studying the relationship between recommendation systems and online polarization.

6.2 Future Scope

Although the project successfully analysed the influence of recommendation systems on user polarization, several improvements and extensions can be explored in the future.

Future Enhancements:

- Use larger and real-time datasets collected directly from social media APIs for more accurate analysis.
- Implement advanced machine learning and deep learning models for improved recommendation and sentiment analysis.
- Analyse additional social media platforms such as:
 - YouTube
 - Instagram

- Facebook
- Incorporate network analysis techniques to study user interaction patterns and community structures.
- Develop hybrid recommendation systems that balance both engagement and diversity.
- Perform real-time recommendation simulations and behavioural analysis.
- Explore methods to reduce algorithmic bias and misinformation spread on social media platforms.

Future research can help create recommendation systems that improve user experience while also promoting balanced information exposure and reducing polarization.

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